Naga Priya Dama

Email: damanagapriya@gmail.com
Mobile: +91 6305658047

Education

- **Jawaharlal Nehru Technological University, Kakinada** — Andhra Pradesh
  Bachelor of Technology - Computer Science and Engineering — *Jul 2018 – Jun 2022*
- **Narayana Jr. College** — Vijayawada, India
  Board of Intermediate Education — *Jun 2016 – Mar 2018*

Experience

- **Software Engineer – Tata Consultancy Services** — Jul 2022 – Present
  - Developed AI-powered enterprise applications using **Google ADK**, Gemini CLI, Claude CLI, Vertex AI, LangGraph, and FastAPI, implementing multi-agent workflows, prompt engineering, AI guardrails, and LLM-as-a-Judge evaluation frameworks.
  - Contributed to enterprise-scale modernization initiatives, including **AI-powered mainframe modernization**, multi-agent software engineering solutions, and AWS-to-GCP cloud migration projects.
  - Designed and developed cloud-native solutions on **Google Cloud Platform (GCP)** using Cloud Functions, Cloud Run, BigQuery, Dataflow, Pub/Sub, Cloud Composer, Workflows, and Managed Instance Groups (MIGs).
  - Built scalable ETL pipelines and backend services using Python, Java Spring Boot, SQL, BigQuery and Cloud Composer to support batch and streaming data processing.
  - Developed CI/CD pipelines and deployment automation using Jenkins, Git, Docker, Linux Shell scripting, Cloud Build, Artifact Registry, and Gitea Actions.
  - Implemented secure, production-grade cloud and AI solutions using REST APIs, cloud-native authentication, AI guardrails, and automated validation frameworks.

Projects

- **AI Mainframe Migration Framework** — Mar 2026 – Present

  Tech Stack: Python, Gemini CLI, Claude CLI, Prompt Engineering, LLM-as-a-Judge, AI Guardrails, Databricks, PySpark

  - Developed the **AIMM Coder** module to automate migration of legacy mainframe artifacts into production-ready **PySpark** code using Gemini CLI, Claude CLI, and advanced prompt engineering techniques.
  - Built the **AIMM Test Generator** module to automatically generate production-grade unit test suites, synthetic test data, execute validation workflows, and measure test coverage for migrated applications.
  - Implemented an **LLM-as-a-Judge** evaluation framework with AI guardrails to validate generated code quality, functional correctness, and migration consistency while minimizing hallucinations.
  - Designed reusable prompt engineering workflows to improve code generation accuracy and automate migration tasks across diverse mainframe workloads.
  - Validated AI-generated PySpark code by executing workloads on Databricks, collaborating with developers to identify migration gaps, refine prompts, and continuously improve framework accuracy, reliability, and generated code quality.
  - Reduced testing cycles by automatically generating robust test suites and synthetic test data, increasing stakeholder confidence that migrated PySpark applications preserve the functional behavior of the original mainframe programs.
- **AI-Powered Data Extraction Pipeline (Internal R&D)** — Jan 2026
  Tech Stack: Python, LangGraph, Azure OpenAI, Streamlit
  - Developed a stateful AI data extraction pipeline using Python, LangGraph, and Azure OpenAI to convert unstructured documents into structured JSON.
  - Implemented conditional workflow orchestration, prompt engineering, and an LLM-as-a-Judge evaluation framework to validate extraction accuracy.

- Built a lightweight Streamlit interface for document processing and visualization of AI-driven extraction results.

- **Cloud-Native Financial AI Agent** Nov 2025 – Feb 2026

  Tech Stack: Python, FastAPI, Google Agent Development Kit (ADK), Azure OpenAI, Azure Web Apps, Azure Blob Storage, Managed Identities, Docker, Gitea Actions

  - Developed a cloud-native conversational AI agent using Python, Google Agent Development Kit (ADK), and Azure OpenAI to analyze and retrieve financial data from enterprise documents.
  - Engineered RESTful APIs using **FastAPI** to expose AI agent capabilities, enabling seamless integration with frontend applications and enterprise services.
  - Implemented secure data retrieval from **Azure Blob Storage**, allowing the AI agent to dynamically discover, access, and analyze financial reports using natural language queries.
  - Integrated **Azure Managed Identities** to provide secure, passwordless authentication for Azure OpenAI, Blob Storage, and other cloud resources.
  - Designed and implemented CI/CD pipelines using **Gitea Actions**, automating containerized application deployment to Azure Web Apps and improving deployment reliability.
  - Containerized the application using Docker and collaborated on building scalable, cloud-native AI solutions following enterprise security and deployment best practices.

- **Multi-Agent Datatype Refactoring System** Sep 2025 – Dec 2025

  Tech Stack: Python, Google ADK, Gemini, Vertex AI, PostgreSQL, Gitea, Git, GitPython, REST APIs, Shell Scripting, GCS

  - Designed and developed a **multi-agent AI system** using Python, Google Agent Development Kit (ADK), and Gemini to automate enterprise-wide datatype refactoring and software remediation across multiple code repositories.
  - Architected a hierarchical agent framework with a **Coordinator Agent** that orchestrated specialized **Database Agent** and **Codebase (Git) Agent** workflows, enabling intelligent task delegation and end-to-end automation.
  - Implemented the **Database Agent** to analyze PostgreSQL schema dependencies, validate datatype changes, automatically generate SQL migration scripts, and upload migration artifacts to Google Cloud Storage (GCS) for review and auditability.
  - Developed the **Codebase Agent** to clone repositories, create feature branches, perform AI-assisted datatype refactoring using Gemini CLI, commit code changes, and automatically generate pull requests through Gitea APIs.
  - Implemented prompt engineering and prompt-injection detection to secure agent interactions, ensuring reliable execution and preventing unauthorized actions during automated workflows.
  - Automated the complete software remediation lifecycle, significantly reducing manual engineering effort while improving consistency, traceability, and developer productivity across large-scale enterprise repositories.

- **AWS to GCP Migration** Apr 2023 – Jul 2025

  Tech Stack: Python, Java, GCP, BigQuery, Dataflow, Cloud Composer, Dataproc, Pub/Sub, Cloud Functions, Jenkins, SQL, Git

  - Contributed to the migration of AWS workloads to GCP for a global business intelligence client, modernizing data processing and analytics infrastructure.
  - Migrated large-scale datasets from **AWS S3 and Redshift to Google Cloud Storage and BigQuery**, ensuring schema compatibility, data integrity, and optimized query performance.
  - Built scalable **ETL pipelines using Dataflow, Pub/Sub, and BigQuery** to support both batch and streaming data ingestion for near real-time analytics.
  - Developed and optimized **BigQuery SQL queries** using window functions, CTEs, and aggregations, reducing processing time by **25%** for large-scale datasets.
  - Converted AWS Glue and Athena jobs into **Cloud Composer (Airflow) DAGs** and orchestrated Dataproc Spark jobs, scheduled workflows, and BigQuery transformations.
  - Designed partitioned and clustered BigQuery tables to improve query performance while reducing storage and compute costs.

- ◦ Built serverless event-driven workflows using **Cloud Functions, Workflows, Pub/Sub, and Cloud Scheduler** for automated data processing.
- ◦ Replaced Cloud Functions with autoscaled **Managed Instance Groups (MIGs)** for compute-intensive workloads, achieving approximately **30% cost savings**.
- ◦ Refactored backend services in **Python and Java Spring Boot** to integrate seamlessly with GCP-native services and cloud APIs.
- ◦ Automated CI/CD pipelines using **Jenkins, Git, Linux Shell scripting**, and maintained deployment pipelines for reliable application delivery.

- **Auth APIs** Nov 2022- March 2023
  Tech Stack: Python, GCP , PostgreSQL,Docker
  - ◦ Developed **RESTful APIs** using **FastAPI** with JWT-based authentication and Swagger UI documentation.
  - ◦ Containerized the application with **Docker and deployed on GCP Cloud Run** via Artifact Registry, ensuring secure, scalable API services.

Certifications

- **Google Generative AI Leader** Oct 2025

  Demonstrated proficiency in Generative AI fundamentals, Gemini, Vertex AI, responsible AI practices, prompt engineering, and enterprise AI adoption on Google Cloud.

- **Google Cloud Associate Cloud Engineer** May 2024
  Focused on deploying, monitoring, and maintaining projects on Google Cloud.

- **AWS Developer Certification** Jul 2024
  Validates foundational understanding of AWS services and cloud architecture.

Skills

- **Programming Languages** Python, Java (Spring Boot), SQL, Shell Scripting
- **Generative AI & Agentic AI** Google ADK, Gemini CLI, Claude CLI, Vertex AI, LangGraph, Prompt Engineering, LLM-as-a-Judge, AI Guardrails, Multi-Agent Systems
- **Google Cloud Platform** BigQuery, Dataflow, Cloud Functions, Cloud Run, Pub/Sub, Workflows, Cloud Composer, Dataproc, GCS, IAM, Managed Instance Groups (MIGs), Cloud Build, Artifact Registry
- **Frameworks & Tools** FastAPI, Docker, Git, Gitea, GitPython, Jenkins, Databricks, Streamlit, Linux
- **Databases** BigQuery, PostgreSQL